## Part 1

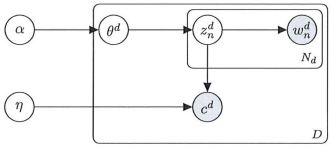


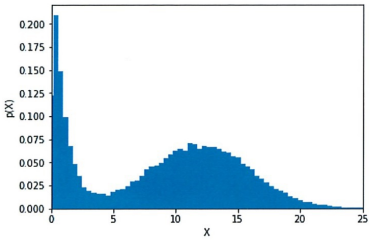

Figure 1: Model A

Consider the graphical model in Figure 1:

**1.** (20%) Write the corresponding factorization of the joint distribution.

$$p(\alpha)p(\eta)\prod_{d=1}^{D} p(\theta^d|\alpha)p(c^d|\eta,\vec{z}^d)\prod_{n=1}^{N} p(z_n^d|\theta^d)\,p(w_n^d|z_n^d)$$

Part 2




**Consider the picture above**

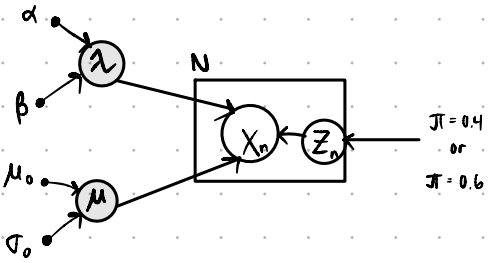
You received a univariate dataset, $\mathcal{D}$ ($\mathcal{D} = \{x_1, x_2, ..., x_N\}$) and plotted its distribution. Your domain knowledge clearly indicates that it contains a mixture of two distributions: 60% of an exponential and 40% normal, defined respectively as Exp($\lambda$) and $\mathcal{N}(\mu, 10)$.

You want to implement a Probabilistic Graphical Model that fits this data. You are essentially interested in inferring the distributions of $\mu$ and $\lambda$.

**1.** (20%) Build the corresponding graphical model

$Exp(\lambda)$

$\mathcal{N}(\mu, 10)$




**2.** (30%) Write the generative process (generative story). Don't forget to make sure to include the appropriate conjugate priors (see last page)

Draw $\mu \sim \mathcal{N}(\mu \mid \mu_0, \sigma_0)$

Draw $\lambda \sim$ Gamma $(\lambda \mid \alpha, \beta)$

For each observation $n \in \{1, ..., N\}$

a) Draw $z_n \sim$ Bernoulli $(z_n \mid \pi = 0.6)$

b) if $z_n = 1$

Draw $x_n \sim Exp(x_n \mid \lambda)$

if $z_n = 0$ :

Draw $x_n \sim \mathcal{N}(x_n \mid \mu, \sigma = 10)$

## Part 3

$$\pi \longrightarrow z \longrightarrow x$$

Consider the following generative process:

1. Draw $\pi \sim \text{Beta}(\pi|2,1)$
2. Draw $z$, such that $z \sim \text{Bernoulli}(z|\pi)$
3. If $z = 0$
   (a) Draw $x$, such that $x \sim \mathcal{N}(x|-2,2)$

   If $z = 1$
   (a) Draw $x$, such that $x \sim \mathcal{N}(x|3,1)$

According to the generative process above, you know that the joint distribution, $p(\pi, z, x)$, factorizes as:

$$p(\pi, z, x) = p(\pi)\, p(z|\pi)\, p(x|z)$$

Based on the generative story and corresponding factorization of the joint distribution:

1. (30%) Formulate the expression for $p(x|\pi)$, using the factorization you have given above. **Simplify your expression as much as possible** (no "$z$" symbol in the final expression).

with all intermediate steps:

$$p(x|\pi) = \frac{p(x,\pi)}{p(\pi)}$$

$$= \frac{\sum_z p(x,z,\pi)}{p(\pi)}$$

$$= \frac{\sum_z \cancel{p(\pi)}\, p(z|\pi)\, p(x|z)}{\cancel{p(\pi)}}$$

$$= \sum_z p(z|\pi)\, p(x|z)$$

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$$p(z=0|\pi)\, p(x|z=0) + p(z=1|\pi)\, p(x|z=1)$$

$$= (1-\pi)\cdot \mathcal{N}(x|-2,2) + \pi \cdot \mathcal{N}(x|3,1) \quad //$$